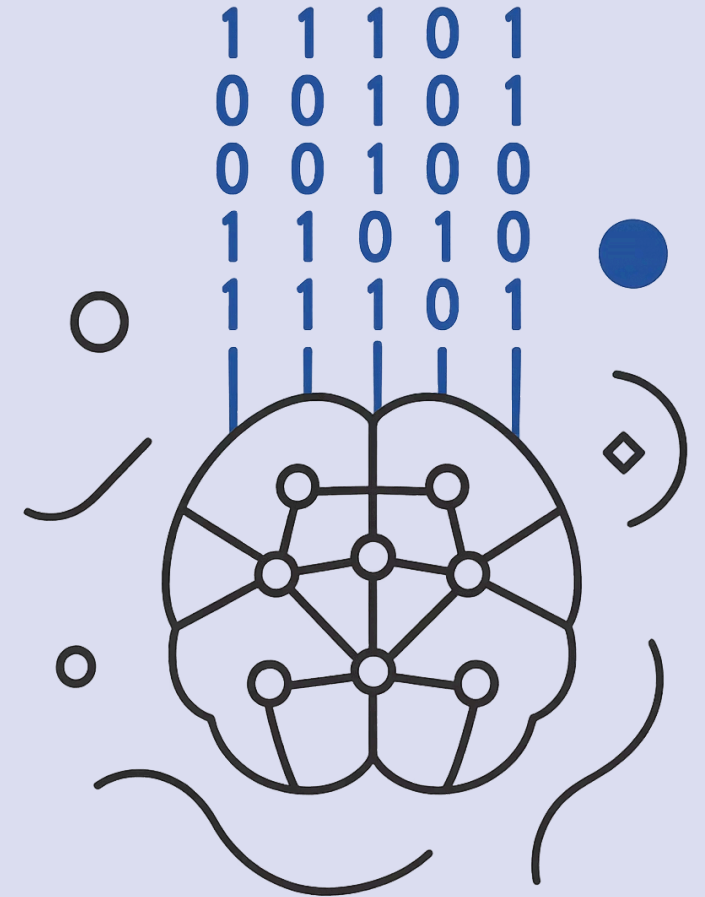


Entering the AI World

From Data to Intelligence

SECTION 1: AI WORLD AND DATA SCIENCE



What We Will Cover

01

Why AI Became Powerful Now

02

The AI Family Tree

03

Where Data Science Fits

04

The Data Science Lifecycle

05

Transition to Hands-On Practice



Did AI Start With ChatGPT?

Many people think AI is a recent invention.

But the dream of making machines intelligent has existed for many decades.



AI did not start with ChatGPT.

What changed recently is that the conditions became right.

The AI Dream Was Old

AI has been researched and attempted for decades.

Earlier AI was held back by fundamental limitations.

1

Less Data

Digital activity was limited

2

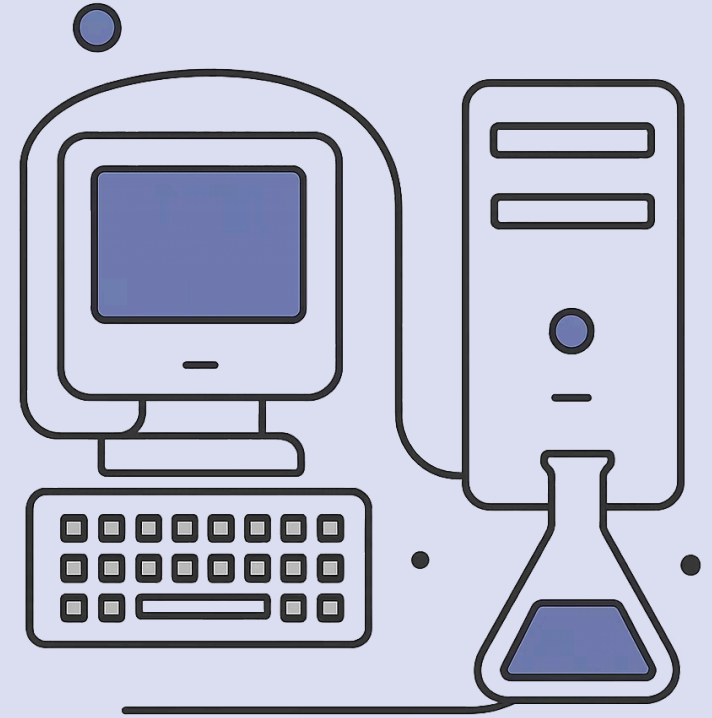
Immature Algorithms

Learning methods were not ready

3

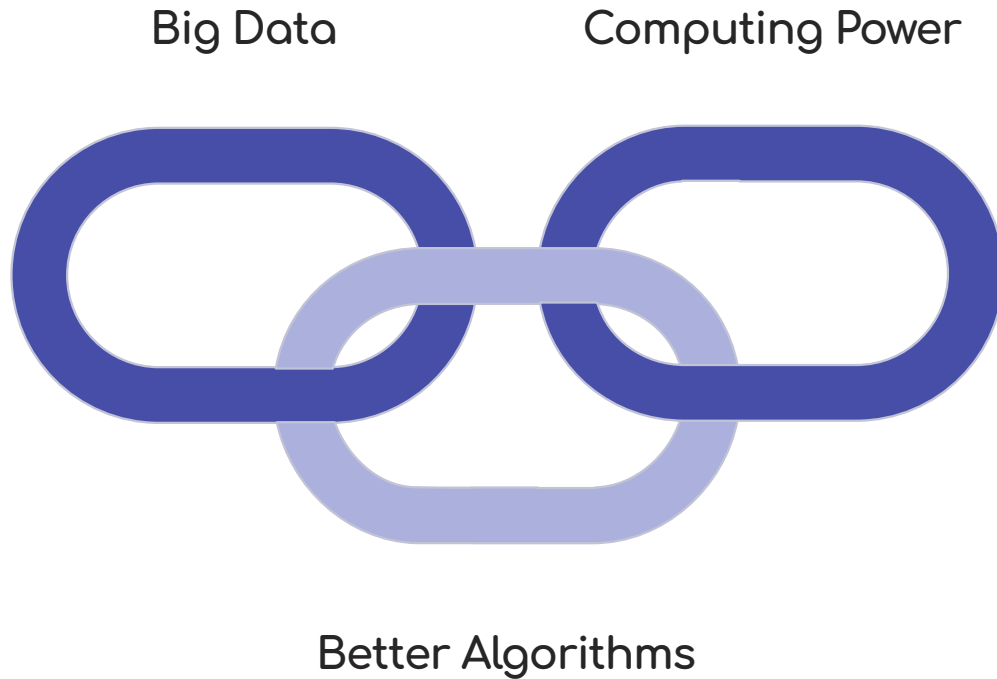
Weak Hardware

Computing was slow and expensive



The Timing Became Right

Modern AI became powerful because three forces matured together.



AI did not become powerful because of one invention.

It became powerful because the full ecosystem became ready.

Force 1: Big Data

Data is the fuel.

Everyday digital activity generates massive examples AI can learn from.

Sources

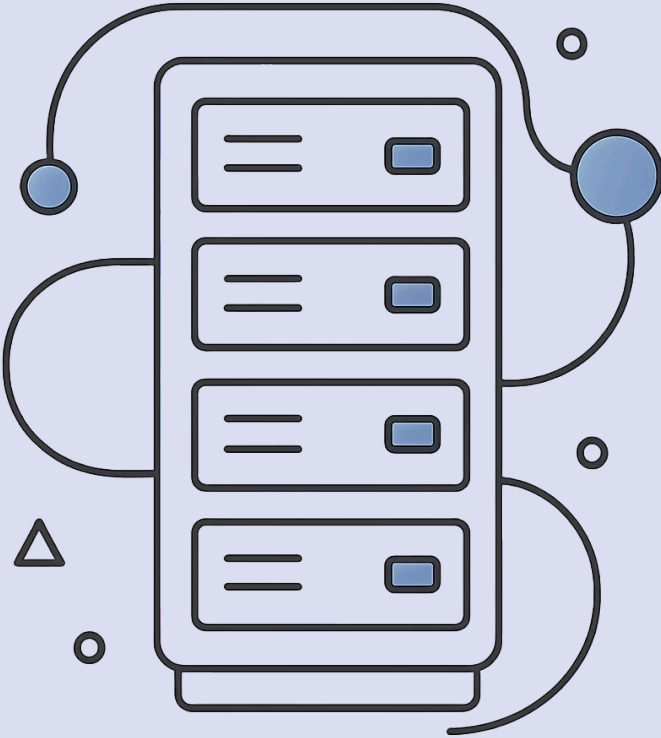
Mobile apps, social media, payments, GPS, sensors, hospitals, video platforms, ...

Why It Matters

More examples reveal better patterns.

Poor data creates poor AI behavior.

❏ Without data, AI has nothing to learn from.



Force 2: Algorithms



Algorithms are the brain.

An algorithm is a step-by-step method used to solve a problem.

In AI, algorithms help machines learn patterns from data.

- Machine Learning
- Deep Learning

Force 3: Computing Power

Computing power is the muscle.

Earlier, large AI experiments were slow and expensive.

Now we have the infrastructure to scale.

- 
- GPUs & TPUs
Specialized AI hardware
 - Cloud Computing
Scalable on-demand resources
 - Distributed Training
Parallel model training at scale
 - Open-Source Tools
Accessible AI frameworks

Why AI Feels Sudden

The public experienced AI directly through chatbots, image generation, and code assistants

but the foundation was built over time.



AI Is Already Around Us

Which of these did you use in the last 24 hours?



Recommendations

Products, videos, courses



Fraud Detection

Banking and payments



Chatbots

Support and assistants



Voice Assistants

Siri, Alexa, Google



Route Suggestions

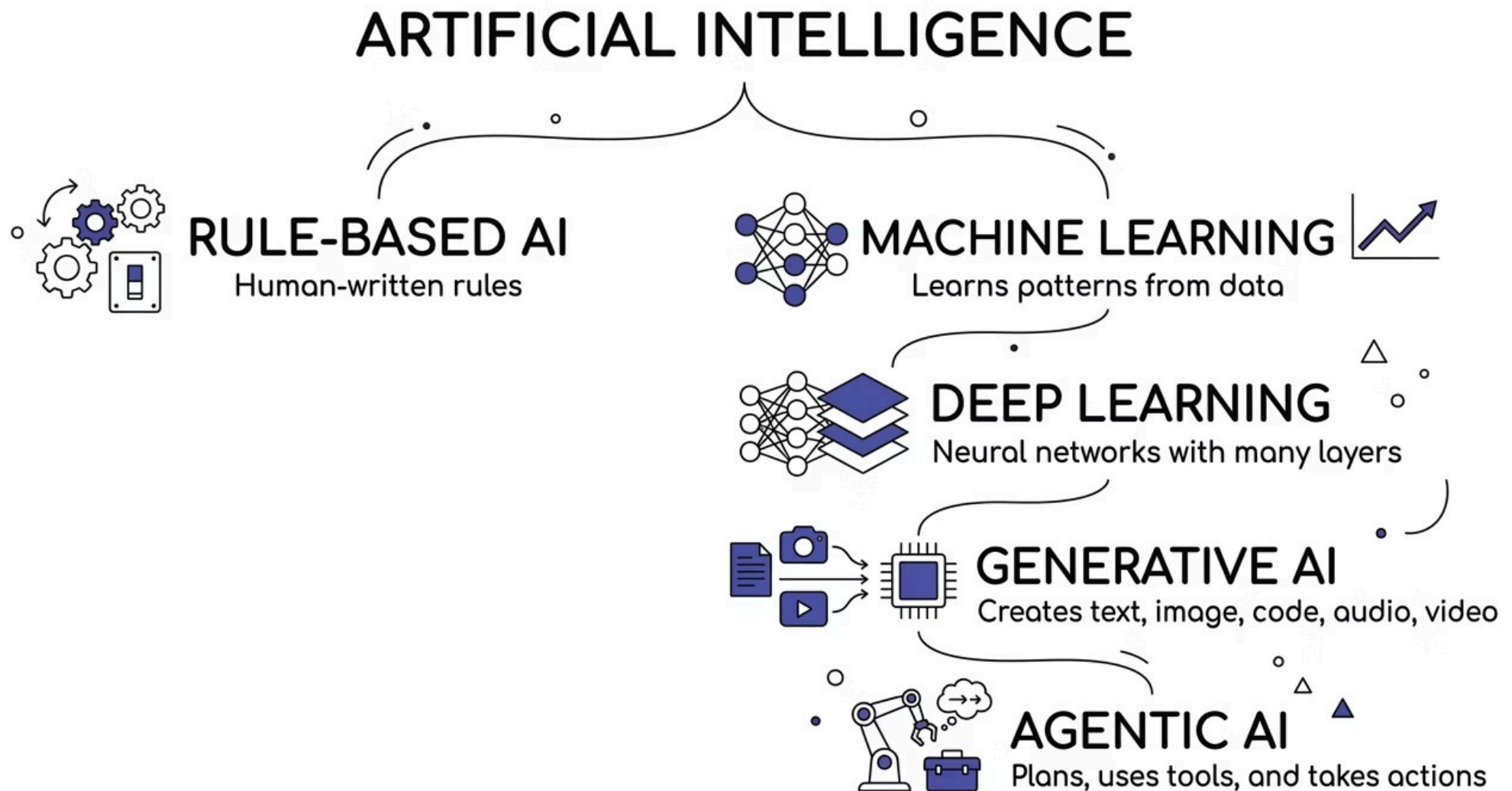
Maps and navigation



Search Ranking

Web and app search

The AI Family Tree



This is a teaching map — not a strict taxonomy. It helps place concepts correctly before going deeper.

Rule-Based AI

How It Works

Humans write every rule explicitly. The machine follows instructions — it does not learn.

```
if marks >= 35:  
    result = "Pass"  
else:  
    result = "Fail"
```

Good For

- Fixed decisions
- Clear business rules
- Compliance checks

Weak For

- Complex patterns
- Image, speech, language
- Ambiguous or changing cases



Machine Learning

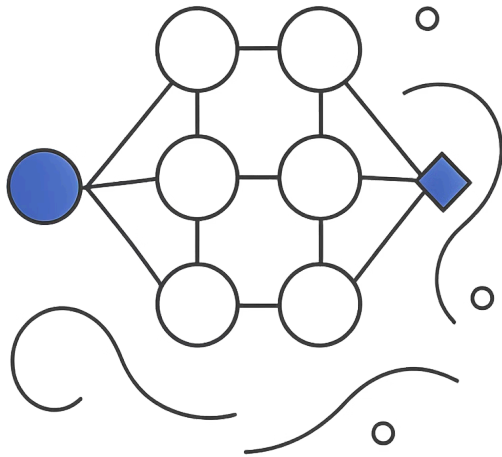
Instead of writing every rule, we give examples.

The algorithm learns which patterns connect to outcomes.

- i Example: Past student data — attendance + assignment score + lab score → the model learns patterns connected to the final outcome.

Good for prediction, classification, pattern recognition, risk scoring, and recommendation.

Deep Learning



Neural Networks With Many Layers

Deep Learning handles complex data types that simpler models struggle with.

- Images and video
- Audio and speech
- Natural language
- Medical scans

❏ Deep Learning became powerful when large data and strong computing power became available.

Generative AI

Generative AI creates new content — not just predictions.

Traditional ML

Predicts a value or category

Example: predict student marks

Generative AI

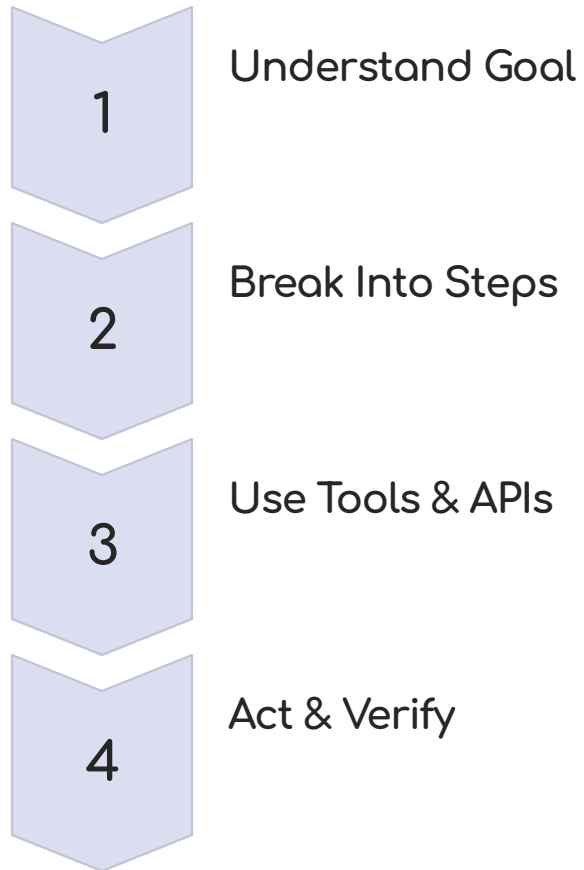
Creates new content

Example: write an explanation, generate an image, produce code

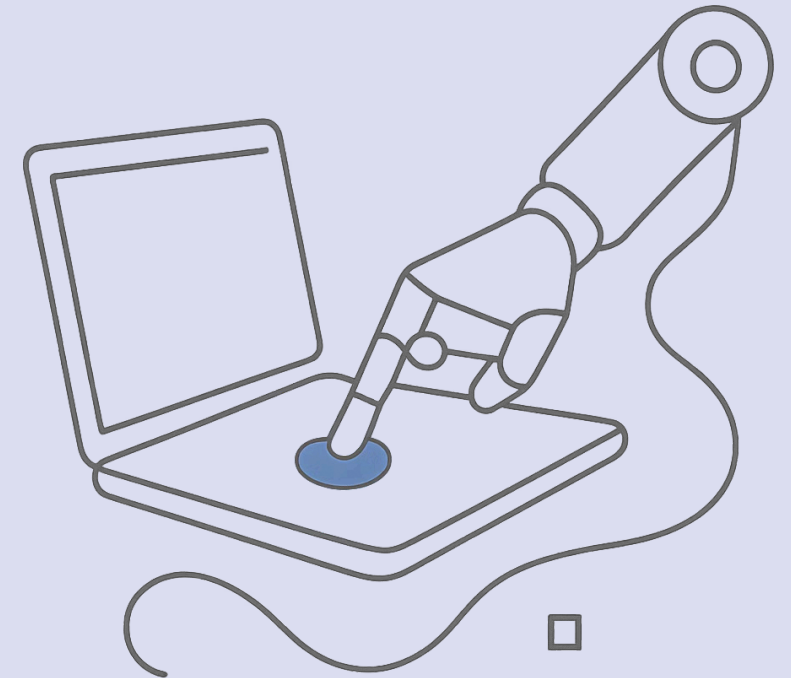
❏ Traditional AI predicts. Generative AI creates.

Agentic AI

Agentic AI moves from answering to acting.



Example: A normal chatbot tells you how to book a ticket. An AI agent searches, compares, confirms, books, and sends a summary.



Data Science vs. AI

Data Science Focuses On

- Understanding and cleaning data
- Exploring patterns
- Statistics and visualization
- Communicating insights
- Supporting decisions

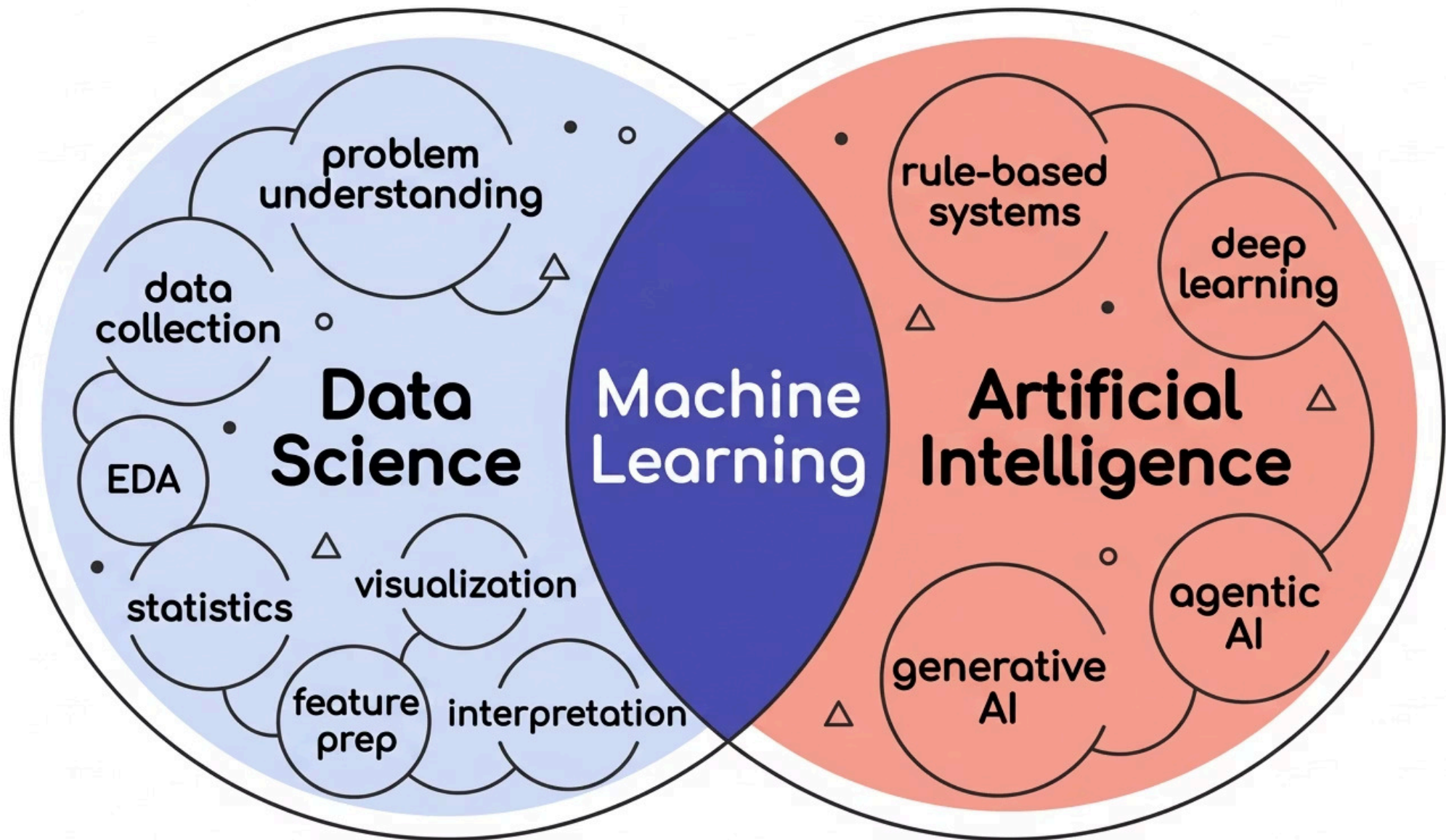
AI Focuses On

- Making machines behave intelligently
- Predicting and understanding
- Generating content
- Planning and acting

 Data Science prepares and extracts value from data.

AI uses data and algorithms to create intelligent behavior.

The Relationship Map



Machine Learning is the major overlap between Data Science and AI — it lives in both worlds.

What Is Data Science?

Data Science uses data to answer questions, find patterns, support decisions, and build data-driven solutions.

It is not only coding.

It spans asking the right question, understanding the domain, cleaning messy data, visualizing clearly, and turning findings into action.

- ❏ Data Science turns raw data into useful decisions.



Model Building Is Only One Part

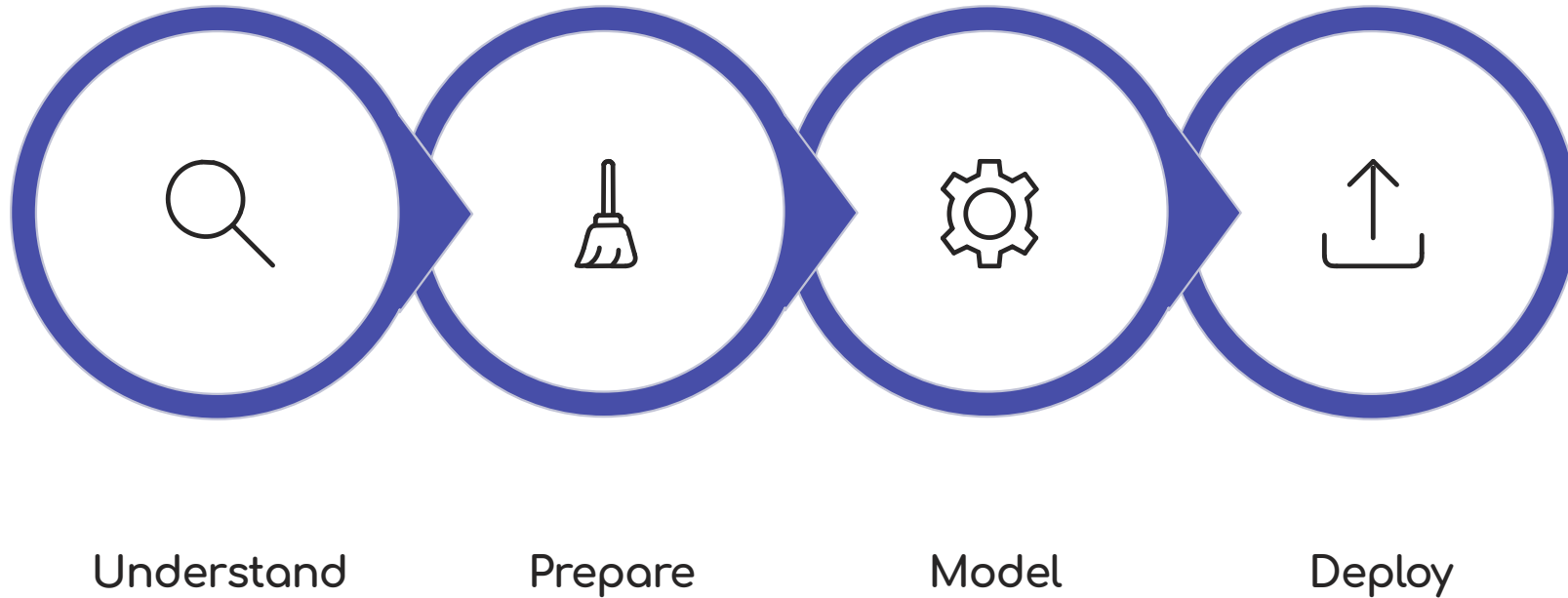
Many beginners think Data Science means only machine learning models.

Real Data Science is broader.

- **Business Understanding**
Define the real problem before touching data
- **Data Quality**
Exploratory analysis and cleaning
- **Storytelling**
Communicating findings clearly
- **Responsible Decisions**
Ethics, fairness, and monitoring

⚠ In real projects, understanding and preparing data often takes more effort than training the model.

The Data Science Lifecycle



Day 1 focuses on the early lifecycle stages. Later sessions move into modeling and deployment.



Step 1: Problem Understanding

Before touching data, define the question clearly.

Weak Question

Analyze student data.

Better Question

Which patterns help identify students who may need early academic support?

☐ A clear problem creates clear analysis.

Step 2: Data Collection

Where Data Comes From

- CSV and Excel files
- Databases and APIs
- Application logs
- Sensors and web platforms
- Survey forms and business systems

Good Collection Asks

- Is the data relevant and reliable?
- Is it legal and ethical to use?
- Does it cover enough examples?
- Are important columns missing?



More data is not always better. Relevant and reliable data is better.

Step 3: Data Cleaning

Real-world data is usually messy. Cleaning is non-negotiable.

Missing Values

Duplicate Records

Wrong Data Types

Inconsistent Spelling

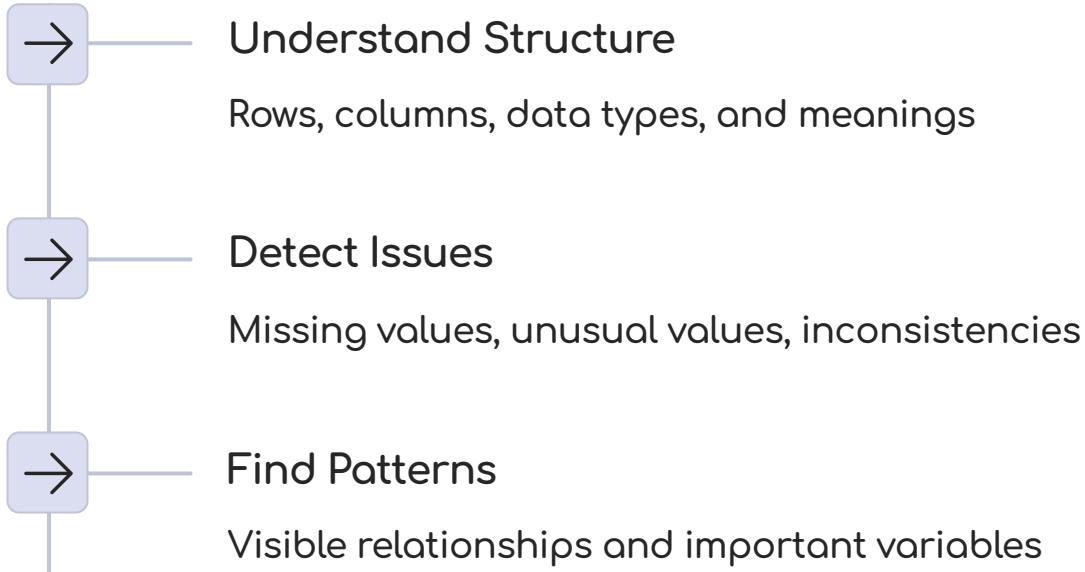
Invalid Values

Mixed Formats

📄 Clean data is the foundation of trustworthy analysis.

Step 4: Exploratory Data Analysis

EDA means listening to the data before asking it to support decisions.



Step 5: Statistics



Key Concepts for Beginners

- Mean, Median, Mode
- Min, Max, Range
- Variance and Standard Deviation
- Correlation
- Distribution

📄 Statistics gives language to data patterns.

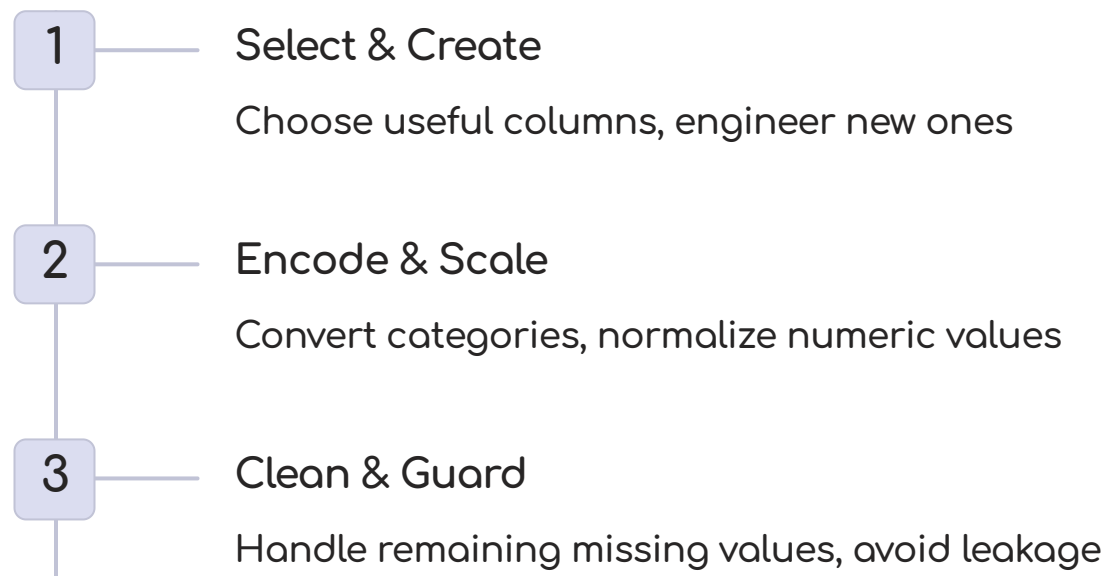
Step 6: Visualization

A good chart makes a pattern easier to see and easier to explain.

Question	Useful Chart
Compare categories	Bar chart
See distribution	Histogram
See spread and outliers	Box plot
Relationship between two numbers	Scatter plot
Trend over time	Line chart
Compare category counts	Count plot

Step 7: Feature Preparation

Feature preparation transforms data into a form that algorithms can use.



□ Clean data is human-readable. Prepared data is model-readable.

Step 8: Modeling

What Modeling Does

Uses algorithms to learn patterns from data for prediction, classification, clustering, anomaly detection, recommendation, or generation.



Not every data science problem needs a model. Sometimes a clear analysis and dashboard are enough.



Step 9: Evaluation

A result is useful only if it is reliable enough for the decision.

For Analysis

- Are insights correct?
- Are charts clear?
- Can stakeholders act on findings?

For Models

- How accurate are predictions?
- Does it work on unseen data?
- Is the model fair and explainable?

Step 10: Deployment and Monitoring

Deployment Forms

- Dashboard or report
- Web app or API
- Automated pipeline
- Model service

Why Monitoring Matters

- Data and user behavior change
- Model performance may drop
- Business rules may shift
- Errors must be caught early

❗ The project is not finished when the model is trained.

It is useful when people can use it safely and reliably.

How Professionals Think

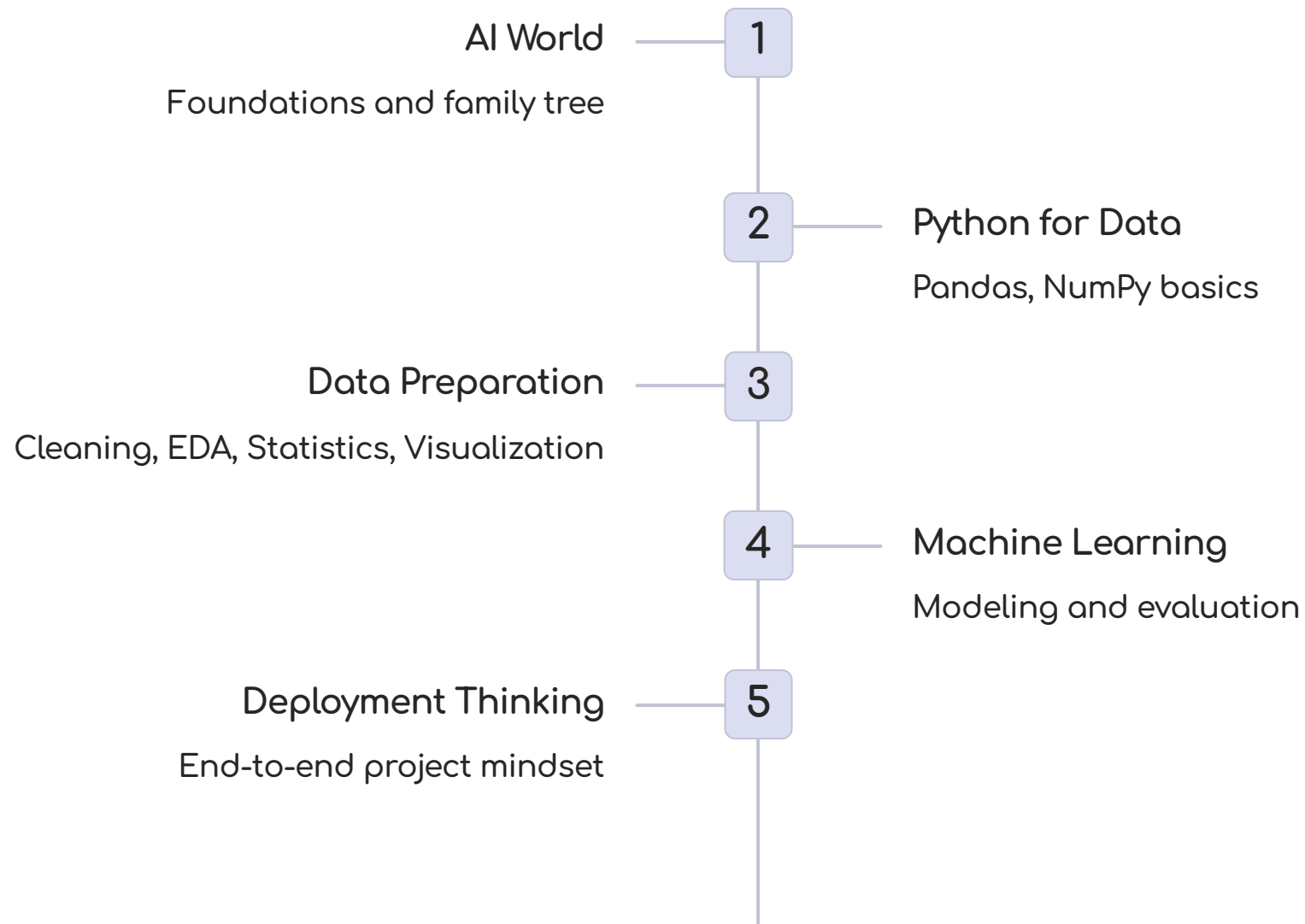
-  Start with the problem, not the tool
-  Understand data before modeling
-  Visualize before concluding
-  Choose metrics based on the real decision
-  Consider ethics, privacy, and fairness

Common Beginner Mistakes

This training will build the habit of asking why before asking how.

- **Skipping Problem Definition**
Starting to code without understanding the question
- **Jumping to ML Too Early**
Ignoring missing values, duplicates, and data quality
- **Trusting Accuracy Blindly**
Not checking what types of mistakes the model makes
- **Explaining Code, Not Insight**
Forgetting the final business decision

Training Roadmap



We will not learn isolated commands.

We will learn how each step connects to an end-to-end data project.



Now We Start Building the Ground

Day 1 builds the foundation. Later days build models on that ground.

Today

AI world, Data Science lifecycle, Python basics, data inspection, cleaning, statistics, visualization

Not Yet

Not full machine learning. Not memorizing syntax. Not isolated commands.



Now that we understand the AI world, we will learn how data becomes the foundation for it.